

AMENDMENT NO. 7 JULY 2008
TO
IS 731 : 1971 SPECIFICATION FOR PORCELAIN
INSULATORS FOR OVERHEAD POWER
LINES WITH A NOMINAL VOLTAGE
GREATER THAN 1 000 V

(*Second Revision*)

[Page 13, *clause 10.1.1(j)*] — Substitute 'Mechanical Performance Test (10.9.1) and Thermal Mechanical Performance Test (only on string insulator units) (10.9.2)' for '24 hours mechanical strength test (for string insulators only when specified by the purchaser) (10.9)'.

(Page 19, *clause 10.9*) — Substitute 'Mechanical Performance Test and Thermal Mechanical Performance Test' for 'Twenty-four Hours Mechanical Test (for String Insulator Units only when Specified by Purchaser)'.

(Page 19, *clause 10.9.1*) — Substitute the following for the existing text:

10.9.1 *Mechanical Performance Test*

This test has an initial stage of mechanical loading and unloading, and a concluding stage of testing the insulator units to failure. The concluding stage is identical to an ordinary electromechanical or mechanical failing load test.

During the initial stage of the test, the insulator units shall be subjected to a tensile load equal to 60 percent of the specified electromechanical or mechanical failing load (whichever is applicable). Unless otherwise agreed, the tensile load shall be applied and immediately removed four times in succession. On the same day, after loading and unloading, the insulator units shall be subjected individually to an electromechanical or mechanical failing load test.

The performance of the insulator units will be determined by a comparison of the failing load values and the fracture pattern obtained during the electromechanical and mechanical failing load test and the same test carried out as the final stage of the mechanical performance test defined here.

Amend No. 7 to IS 731 : 1971

NOTES

1 The test may be more decisive if the 60 percent load is applied and removed more than four times in succession.

2 The insulator units may be coupled together in series and/or in parallel when subjected to the 60 percent load. When parallel coupled, the insulator units must be equally loaded.

3 The test may not give information on the internally stressed zone If the failure occurs in a metal part such as the socket or clevis of a cap or an insulator pin. In such cases, this test is not suitable for a sample test, but by agreement it may be used as a design test or special qualification test. It is possible to investigate the fundamental insulator design by using metal parts suitably strengthened so that failure occurs in the internally stressed zone of the insulator.

Precautions should be taken so that strengthening the metal parts does not affect the fundamental stress relation.

4 Voltage may be applied to the insulator units (Type B only) whenever it is desired in the course of the test to detect mechanical failure of the insulating part (puncture).

(Page 19, clause **10.9.1**, last line) — Insert the following and renumber **10.9.2** as **10.9.3**:

10.9.2 Thermal-Mechanical Performance Test

During the initial stage of the test, the insulator units shall be subjected to four 24-hours cycles of cooling and heating, and a tensile load equal to 60 percent of the specified electro-mechanical or mechanical falling load. The tensile load shall be applied to the insulator units at room temperature before starting the first thermal cycle.

Unless otherwise agreed, each 24-hour cycle shall comprise a cooling to $-30 \pm 5^{\circ}\text{C}$ and a heating to $+40 \pm 5^{\circ}\text{C}$. The temperature figure refers to the surrounding air. The temperature sequence shall be first cooling, then heating. The test equipment shall be such as to permit the minimum and maximum temperatures each to be kept during atleast four consecutive hours of the temperature cycle.

The tensile load shall be completely removed and re-applied towards the end of each heating period and then last one accepted.

On completion of the fourth 24-hour cycle and cooling to room temperature, the tensile load shall be removed. On the same day, after this load removal, the

Amend No. 7 to IS 731 : 1971

insulator unit shall be subjected individually to an electromechanical or mechanical failing load test.

The performance of the insulator units will be determined by a comparison of the failing load values and the fracture pattern obtained during the electromechanical and mechanical failing load test and the same test carried out as the final stage of the mechanical performance test defined here.

Further the tolerances on the temperatures of the hot and cold cycles shall be respected in such a way so as to ensure a minimum difference of 70 degrees between the recorded hot and cold temperatures.

NOTES

1 This thermal-mechanical performance test has reference to the fundamental insulator design in respect of the internal stresses, and should not be repeated on types which differ in outward form only, that is, the disc of the insulating part or the coupling ends of the metal fittings. Changes in internal design or in manufacturing processes are reasons for re-testing. The test may not give information on the internally stressed zone if the failure occurs in a metal part such as the socket or clevis of a cap or an insulator pin. In such cases, it is possible to investigate the fundamental insulator design by using metal parts suitably strengthened so that failure occurs in the internally stressed zone of the insulator.

Precautions should be taken so that strengthening the metal parts does not affect the fundamental stress relation.

2 The insulator units may be coupled together in series and/or in parallel when subjected to the thermal cycles and the 60 percent load. When parallel coupled, the insulator units must be equally loaded.

3 Loose coupling pins, for example, those used with insulators of the long rod type, should not be included in the mechanical test since they are not part of the internal design of the insulator.

4 Voltage may be applied to the insulator units (Type B only) whenever desired in the course of the test, but without altering the test cycle, in order to detect mechanical failure of the insulating part (puncture).'

(Page 21, clause **10.14.2**) — Substitute '50 percent of the specified electro-mechanical failing load' for '40 percent of the specified electro-mechanical failing load'.

(ET 06)